

	$\mathcal{K}_0^{\#1+}$	$\mathcal{K}_0^{\#1-}$					
	$\mathcal{K}_0^{\#1+} \dagger$	$\frac{1}{2}(-3\Upsilon_1 k^2 - 3\binom{(2)}{\mathcal{K}_3})$	0				
$\mathcal{K}_0^{\#1+} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{K}_1^{\#1-}{}_{\alpha\beta}$	$\mathcal{K}_1^{\#2-}{}_{\alpha\beta}$	$\mathcal{K}_1^{\#1-}{}_{\alpha}$	$\mathcal{K}_1^{\#2-}{}_{\alpha}$	
	$\mathcal{K}_1^{\#1+} \dagger^{\alpha\beta}$	$\Upsilon_2 k^2$	$-\frac{\Upsilon_3 k^2}{2\sqrt{2}}$	0	0		
	$\mathcal{K}_1^{\#2+} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_3 k^2}{2\sqrt{2}}$	$-\frac{\Upsilon_4 k^2}{4}$	0	0		
	$\mathcal{K}_1^{\#1+} \dagger^{\alpha}$	0	0	$\frac{1}{2}(\Upsilon_5 k^2 - 2\binom{(2)}{\mathcal{K}_3})$	$\frac{-\Upsilon_5 k^2 + 2\binom{(2)}{\mathcal{K}_3}}{2\sqrt{2}}$		
	$\mathcal{K}_1^{\#2+} \dagger^{\alpha}$	0	0	$\frac{-\Upsilon_5 k^2 + 2\binom{(2)}{\mathcal{K}_3}}{2\sqrt{2}}$	$\frac{1}{4}(\Upsilon_5 k^2 - 2\binom{(2)}{\mathcal{K}_3})$	$\mathcal{K}_2^{\#1-}{}_{\alpha\beta}$	$\mathcal{K}_2^{\#1-}{}_{\alpha\beta\chi}$
					$\mathcal{K}_2^{\#1+} \dagger^{\alpha\beta}$	0	0
					$\mathcal{K}_2^{\#1+} \dagger^{\alpha\beta\chi}$	0	$\frac{3k^2\binom{(4)}{\mathcal{K}_{15}}}{2}$

	$\mathcal{T}_{0^+}^{\#1}$	$\mathcal{T}_{0^- \alpha\beta\chi}^{\#1}$				
$\mathcal{T}_{0^+}^{\#1} \dagger$	$\frac{1}{\text{Det}(0^+)}$	0				
$\mathcal{T}_{0^-}^{\#1} \dagger^{\alpha\beta\chi}$	0	0	$\mathcal{T}_{1^+ \alpha\beta}^{\#1}$	$\mathcal{T}_{1^+ \alpha\beta}^{\#2}$	$\mathcal{T}_{1^- \alpha}^{\#1}$	$\mathcal{T}_{1^- \alpha}^{\#2}$
$\mathcal{T}_{1^+}^{\#1} \dagger^{\alpha\beta}$	$-\frac{\Upsilon_4 k^2}{4 \text{Det}(1^+)}$	$\frac{\Upsilon_3 k^2}{2 \sqrt{2} \text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#2} \dagger^{\alpha\beta}$	$\frac{\Upsilon_3 k^2}{2 \sqrt{2} \text{Det}(1^+)}$	$\frac{\Upsilon_2 k^2}{\text{Det}(1^+)}$	0	0		
$\mathcal{T}_{1^+}^{\#1} \dagger^\alpha$	0	0	$\frac{2}{3 \text{Det}(1^-)}$	$-\frac{\sqrt{2}}{3 \text{Det}(1^-)}$		
$\mathcal{T}_{1^+}^{\#2} \dagger^\alpha$	0	0	$-\frac{\sqrt{2}}{3 \text{Det}(1^-)}$	$\frac{1}{3 \text{Det}(1^-)}$	$\mathcal{T}_{2^+ \alpha\beta}^{\#1}$	$\mathcal{T}_{2^- \alpha\beta\chi}^{\#1}$
	$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta}$	0	0			
	$\mathcal{T}_{2^+}^{\#1} \dagger^{\alpha\beta\chi}$	0	$\frac{2}{3 k^2 \langle k_1 \rangle}$			

$$\begin{aligned} & \text{Abbreviations used in matrices} \\ Y_1 &= \binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} \& Y_2 = 2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_3} \& Y_3 = 2 \binom{(4)}{\kappa_{15}} - \binom{(4)}{\kappa_4} \& \\ Y_4 &= 5 \binom{(4)}{\kappa_{15}} - 2 \left(\binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4} \right) \& Y_5 = 3 \binom{(4)}{\kappa_{15}} - 2 \binom{(4)}{\kappa_2} \& \text{Det}(0^+) = -\frac{3}{2} (k^2 \left(\binom{(4)}{\kappa_1} + \binom{(4)}{\kappa_2} \right) + \binom{(2)}{\kappa_3}) \& \\ \text{Det}(1^+) &= -\frac{1}{8} k^4 (24 \binom{(4)}{\kappa_{15}}^2 + (2 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4})^2 - 6 \binom{(4)}{\kappa_{15}} (3 \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4})) \end{aligned}$$

$$\begin{aligned} & \text{Lagrangian} \\ & \binom{2}{K} 3 \mathcal{K}^{\alpha \beta} \mathcal{K}^{\chi}_{\beta \chi} + \binom{4}{K} 1 \partial_{\beta} \mathcal{K}^{\delta}_{\chi \delta} \partial^{\chi} \mathcal{K}^{\alpha \beta}_{\alpha} + \binom{4}{K} 2 \partial_{\chi} \mathcal{K}^{\delta}_{\beta \delta} \partial^{\chi} \mathcal{K}^{\alpha \beta}_{\alpha} + \binom{4}{K} 3 \partial_{\beta} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\alpha \chi} + \binom{4}{K} 4 \partial_{\alpha} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\beta \chi} + \\ & 3 \binom{4}{K} 15 \partial_{\beta} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\chi \alpha} - \binom{4}{K} 3 \partial_{\beta} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\chi \alpha} - 3 \binom{4}{K} 15 \partial^{\chi} \mathcal{K}^{\alpha \beta}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\chi \beta} - \frac{9}{4} \binom{4}{K} 15 \partial_{\alpha} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\beta \chi} + \\ & \frac{1}{2} \binom{4}{K} 3 \partial_{\alpha} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\beta \chi} + \frac{1}{2} \binom{4}{K} 4 \partial_{\alpha} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}_{\beta \chi} + \binom{4}{K} 15 \partial_{\delta} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial^{\delta} \mathcal{K}^{\delta}_{\beta \alpha \chi} + \binom{4}{K} 15 \partial_{\delta} \mathcal{K}^{\alpha \beta \chi}_{\alpha} \partial^{\delta} \mathcal{K}^{\delta}_{\beta \alpha \chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{T}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{T}_1^{\#1\alpha} + 2 \mathcal{T}_1^{\#2\alpha} = 0$	3	$\partial_\chi \partial^\alpha \mathcal{T}^\beta_{\beta\chi} + \partial_\chi \partial^\chi \mathcal{T}^{\beta\alpha}_{\beta} = 3 \partial_\chi \partial_\beta \mathcal{T}^{\beta\alpha\chi}$	
$\mathcal{T}_2^{\#1\alpha\beta} = 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{T}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{T}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial_\delta \mathcal{T}^{\chi\delta}_{\chi} = 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{T}^{\chi}_{\chi} + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{T}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{T}^{\beta\alpha\chi})$	
Total # constraint(s):	8		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{-189 \kappa_{15}^{(4)2} + 54 \kappa_{15}^{(4)} (2 \kappa_3^{(4)} + \kappa_4^{(4)}) - 4 (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2}{\kappa_{15}^{(4)} (24 \kappa_{15}^{(4)2} + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 - 6 \kappa_{15}^{(4)} (3 \kappa_3^{(4)} + 2 \kappa_4^{(4)}))}$

	1	0	$\sqrt{s}$	$ \begin{aligned} & 5878656 \overset{(4)}{K}_{15}^8 - 11057472 \overset{(4)}{K}_{15}^7 \overset{(4)}{K}_3 + 9558648 \overset{(4)}{K}_{15}^6 \overset{(4)}{K}_3^2 - 4875552 \overset{(4)}{K}_{15}^5 \overset{(4)}{K}_3^3 + 1559088 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^4 - \\ & 295488 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^5 + 25920 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^6 - 6998400 \overset{(4)}{K}_{15}^7 \overset{(4)}{K}_4 + 11220768 \overset{(4)}{K}_{15}^6 \overset{(4)}{K}_3 \overset{(4)}{K}_4 - \\ & 8211456 \overset{(4)}{K}_{15}^5 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4 + 3386448 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4 - 777600 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4 + 77760 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^5 \overset{(4)}{K}_4 + \\ & 3312576 \overset{(4)}{K}_{15}^6 \overset{(4)}{K}_4^2 - 4589784 \overset{(4)}{K}_{15}^5 \overset{(4)}{K}_3 \overset{(4)}{K}_4^2 + 2755620 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^2 - \\ & 816480 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4^2 + 97200 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^2 - 851472 \overset{(4)}{K}_{15}^5 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4 + 995328 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3 \overset{(4)}{K}_4^3 - \\ & 427680 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^3 + 64800 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4^3 + 134622 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4 - 111780 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^4 + \\ & 24300 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^4 - 11664 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4 + 4860 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^5 \overset{(4)}{K}_4 + 405 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^6 \overset{(4)}{K}_4 + \\ & (7776 \overset{(4)}{K}_{15}^6 \overset{(4)}{K}_3^5 \overset{(4)}{K}_4 - 648 \overset{(4)}{K}_{15}^5 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4 - 12960 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4 + 12096 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4 - 4320 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^5 \overset{(4)}{K}_4 + 576 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^5 \overset{(4)}{K}_4 + \\ & 41472 \overset{(4)}{K}_{15}^5 \overset{(4)}{K}_4^4 - 63828 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3 \overset{(4)}{K}_4^4 + 40176 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^4 - 12096 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4^4 + 1440 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^4 - \\ & 34020 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^2 + 33372 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3 \overset{(4)}{K}_4^2 - 11664 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^2 + 1440 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4^2 + 8154 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^3 - \\ & 4752 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3 \overset{(4)}{K}_4^3 + 720 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^3 - 702 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4 + 180 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3 \overset{(4)}{K}_4^4 + 18 \overset{(4)}{K}_{15}^5 \overset{(4)}{K}_4^5) \#1 + \\ & (-7137 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3 + 7452 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4 - 3468 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^2 + 864 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4 - 112 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^4 + 6300 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4 - \\ & 4764 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3 \overset{(4)}{K}_4^4 + 1608 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4 - 224 \overset{(4)}{K}_{15}^3 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4 - 2010 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4 + \\ & 960 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^2 - 168 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^2 + 186 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4^4 - 56 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^4 - 7 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_4^4) \#1^2 + \\ & (-48 \overset{(4)}{K}_{15}^2 \overset{(4)}{K}_3 \overset{(4)}{K}_4^2 + 36 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^2 \overset{(4)}{K}_4^2 - 8 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^3 \overset{(4)}{K}_4^2 + 24 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^2 - 8 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_3^4 \overset{(4)}{K}_4^2 - 2 \overset{(4)}{K}_{15}^4 \overset{(4)}{K}_4^2) \#1^3 + \#1^4 \&, 1] / \\ & (\overset{(4)}{K}_{15} (24 \overset{(4)}{K}_{15}^2 + (2 \overset{(4)}{K}_3 + \overset{(4)}{K}_4)^2 - 6 \overset{(4)}{K}_{15} (3 \overset{(4)}{K}_3 + 2 \overset{(4)}{K}_4))) \end{aligned} $
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	1	0	$ \begin{aligned} & \text{Root[} \\ & 5878656 \binom{(4)}{K_{15}}^8 - 11057472 \binom{(4)}{K_{15}}^7 \binom{(4)}{K_3} + 9558648 \binom{(4)}{K_{15}}^6 \binom{(4)}{K_3}^2 - 4875552 \binom{(4)}{K_{15}}^5 \binom{(4)}{K_3}^3 + 1559088 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_3}^4 - \\ & 295488 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3}^5 + 25920 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3}^6 - 6998400 \binom{(4)}{K_{15}}^7 \binom{(4)}{K_4} + 11220768 \binom{(4)}{K_{15}}^6 \binom{(4)}{K_3} \binom{(4)}{K_4} - \\ & 8211456 \binom{(4)}{K_{15}}^5 \binom{(4)}{K_3}^2 \binom{(4)}{K_4} + 3386448 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_3} \binom{(4)}{K_4}^3 - 777600 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3} \binom{(4)}{K_4}^4 + 77760 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3}^2 \binom{(4)}{K_4}^5 \binom{(4)}{K_4} + \\ & 3312576 \binom{(4)}{K_{15}}^6 \binom{(4)}{K_4}^2 - 4589784 \binom{(4)}{K_{15}}^5 \binom{(4)}{K_3} \binom{(4)}{K_4}^2 + 2755620 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \binom{(4)}{K_4}^2 - \\ & 816480 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3} \binom{(4)}{K_4}^3 \binom{(4)}{K_4}^2 + 97200 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} \binom{(4)}{K_4}^4 \binom{(4)}{K_4}^2 - 851472 \binom{(4)}{K_{15}}^5 \binom{(4)}{K_4}^3 + 995328 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_3} \binom{(4)}{K_4}^3 - \\ & 427680 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \binom{(4)}{K_4}^3 + 64800 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} \binom{(4)}{K_4}^3 \binom{(4)}{K_4}^3 + 134622 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_4}^4 - 111780 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3} \binom{(4)}{K_4}^4 + \\ & 24300 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3}^2 \binom{(4)}{K_4}^4 - 11664 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_4}^5 + 4860 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} \binom{(4)}{K_4}^5 + 405 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_4}^6 + \\ & (7776 \binom{(4)}{K_{15}}^6 - 648 \binom{(4)}{K_{15}}^5 \binom{(4)}{K_3} - 12960 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_3}^2 + 12096 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3}^3 - 4320 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3}^4 + 576 \binom{(4)}{K_{15}} \binom{(4)}{K_3}^5 + \\ & 41472 \binom{(4)}{K_{15}}^5 \binom{(4)}{K_4} - 63828 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_3} \binom{(4)}{K_4} + 40176 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \binom{(4)}{K_4} - 12096 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} \binom{(4)}{K_4}^3 \binom{(4)}{K_4} + 1440 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 \binom{(4)}{K_4} - \\ & 34020 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_4}^2 + 33372 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3} \binom{(4)}{K_4}^2 - 11664 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \binom{(4)}{K_4}^2 + 1440 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 \binom{(4)}{K_4}^2 + 8154 \binom{(4)}{K_{15}} \binom{(4)}{K_4}^3 \binom{(4)}{K_4}^3 - \\ & 4752 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} \binom{(4)}{K_4}^3 + 720 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 - 702 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_4}^4 + 180 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 + 18 \binom{(4)}{K_{15}} \binom{(4)}{K_4}^5 \#1 + \\ & (-7137 \binom{(4)}{K_{15}}^4 + 7452 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_3} - 3468 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3}^2 + 864 \binom{(4)}{K_{15}} \binom{(4)}{K_3}^3 - 112 \binom{(4)}{K_{15}}^4 \binom{(4)}{K_4} + 6300 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4} - \\ & 4764 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} \binom{(4)}{K_4} + 1608 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \binom{(4)}{K_4} - 224 \binom{(4)}{K_{15}}^3 \binom{(4)}{K_4}^3 - 2010 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} \binom{(4)}{K_4}^2 + \\ & 960 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 - 168 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_4}^2 + 186 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 - 56 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 - 7 \binom{(4)}{K_{15}} \binom{(4)}{K_4}^4 \#1^2 + \\ & (-48 \binom{(4)}{K_{15}}^2 + 36 \binom{(4)}{K_{15}} \binom{(4)}{K_3} - 8 \binom{(4)}{K_{15}}^2 \binom{(4)}{K_3} + 24 \binom{(4)}{K_{15}} \binom{(4)}{K_4} - 8 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4} - 2 \binom{(4)}{K_{15}} \binom{(4)}{K_4}^2 \#1^3 + \#1^4 \&, 2] / \\ & (\binom{(4)}{K_{15}} (24 \binom{(4)}{K_{15}} + (2 \binom{(4)}{K_3} + \binom{(4)}{K_4})^2 - 6 \binom{(4)}{K_{15}} (3 \binom{(4)}{K_3} + 2 \binom{(4)}{K_4}))) \end{aligned} $
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	1	0	$ \begin{aligned} & \text{Root[} \\ & 5878656 \binom{(4)}{K_{15}} \binom{8}{-11057472 \binom{(4)}{K_{15}} \binom{7}{K_3} \binom{(4)}{K_3} + 9558648 \binom{(4)}{K_{15}} \binom{6}{K_3} \binom{(4)}{K_3}^2 - 4875552 \binom{(4)}{K_{15}} \binom{5}{K_3} \binom{(4)}{K_3}^3 + 1559088 \binom{(4)}{K_{15}} \binom{4}{K_3} \binom{(4)}{K_3}^4 - \\ & 295488 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3}^5 + 25920 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3}^6 - 6998400 \binom{(4)}{K_{15}} \binom{7}{K_4} \binom{(4)}{K_4} + 11220768 \binom{(4)}{K_{15}} \binom{6}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4} - \\ & 8211456 \binom{(4)}{K_{15}} \binom{5}{K_3} \binom{(4)}{K_3}^2 \binom{(4)}{K_4} + 3386448 \binom{(4)}{K_{15}} \binom{4}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4} - 777600 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 + 77760 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3}^5 \binom{(4)}{K_4} + \\ & 3312576 \binom{(4)}{K_{15}} \binom{6}{K_4} \binom{(4)}{K_4}^2 - 4589784 \binom{(4)}{K_{15}} \binom{5}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 + 2755620 \binom{(4)}{K_{15}} \binom{4}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \binom{(4)}{K_4} - \\ & 816480 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 + 97200 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 \binom{(4)}{K_4}^2 - 851472 \binom{(4)}{K_{15}} \binom{5}{K_4} \binom{(4)}{K_4}^3 + 995328 \binom{(4)}{K_{15}} \binom{4}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 - \\ & 427680 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 + 64800 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 + 134622 \binom{(4)}{K_{15}} \binom{4}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 - 111780 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 + \\ & 24300 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 - 11664 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^5 + 4860 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^5 + 405 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_4}^6 + \\ & (7776 \binom{(4)}{K_{15}} \binom{6}{K_4} - 648 \binom{(4)}{K_{15}} \binom{5}{K_3} \binom{(4)}{K_3} - 12960 \binom{(4)}{K_{15}} \binom{4}{K_3} \binom{(4)}{K_3}^2 + 12096 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3}^3 - 4320 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3}^4 + 576 \binom{(4)}{K_{15}} \binom{(4)}{K_3}^5 + \\ & 41472 \binom{(4)}{K_{15}} \binom{5}{K_4} \binom{(4)}{K_4} - 63828 \binom{(4)}{K_{15}} \binom{4}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4} + 40176 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 - 12096 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 + 1440 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 \binom{(4)}{K_4} - \\ & 34020 \binom{(4)}{K_{15}} \binom{4}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 + 33372 \binom{(4)}{K_{15}} \binom{3}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 - 11664 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 + 1440 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \binom{(4)}{K_4}^2 + 8154 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 \binom{(4)}{K_4} - \\ & 4752 \binom{(4)}{K_{15}} \binom{2}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 + 720 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \binom{(4)}{K_4}^3 - 702 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 + 180 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 + 18 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4}^5 \#1 + \\ & (-7137 \binom{(4)}{K_{15}} + 7452 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} - 3468 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3}^2 + 864 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3}^3 - 112 \binom{(4)}{K_{15}} \binom{(4)}{K_3}^4 + 6300 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_4} - \\ & 4764 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4} + 1608 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 - 224 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 - 2010 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 + \\ & 960 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 - 168 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 + 186 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^3 - 56 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^4 - 7 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^5 \#1^2 + \\ & (-48 \binom{(4)}{K_{15}}^2 + 36 \binom{(4)}{K_{15}} \binom{(4)}{K_3} - 8 \binom{(4)}{K_{15}} \binom{(4)}{K_3}^2 + 24 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} - 8 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 - 2 \binom{(4)}{K_{15}} \binom{(4)}{K_3} \binom{(4)}{K_3} \binom{(4)}{K_4}^2 \#1^3 + \#1^4 \&, 3] / \\ & (\binom{(4)}{K_{15}} (24 \binom{(4)}{K_{15}} + (2 \binom{(4)}{K_3} + \binom{(4)}{K_4})^2 - 6 \binom{(4)}{K_{15}} (3 \binom{(4)}{K_3} + 2 \binom{(4)}{K_4}))) \end{aligned} $
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	1	0	$ \begin{aligned} & \text{Root[} \\ & 5878656 \kappa_{15}^{(4)8} - 11057472 \kappa_{15}^{(4)7} \kappa_3^{(4)} + 9558648 \kappa_{15}^{(4)6} \kappa_3^{(4)2} - 4875552 \kappa_{15}^{(4)5} \kappa_3^{(4)3} + 1559088 \kappa_{15}^{(4)4} \kappa_3^{(4)4} - \\ & 295488 \kappa_{15}^{(4)3} \kappa_3^{(4)5} + 25920 \kappa_{15}^{(4)2} \kappa_3^{(4)6} - 6998400 \kappa_{15}^{(4)} \kappa_3^{(4)7} + 11220768 \kappa_{15}^{(4)} \kappa_3^{(4)} \kappa_4^{(4)} - \\ & 8211456 \kappa_{15}^{(4)5} \kappa_3^{(4)2} \kappa_4^{(4)} + 3386448 \kappa_{15}^{(4)4} \kappa_3^{(4)3} \kappa_4^{(4)} - 777600 \kappa_{15}^{(4)3} \kappa_3^{(4)4} \kappa_4^{(4)} + 77760 \kappa_{15}^{(4)2} \kappa_3^{(4)5} \kappa_4^{(4)} + \\ & 3312576 \kappa_{15}^{(4)6} \kappa_4^{(4)2} - 4589784 \kappa_{15}^{(4)5} \kappa_3^{(4)} \kappa_4^{(4)2} + 2755620 \kappa_{15}^{(4)4} \kappa_3^{(4)} \kappa_4^{(4)2} \kappa_4^{(4)2} - \\ & 816480 \kappa_{15}^{(4)3} \kappa_3^{(4)3} \kappa_4^{(4)2} + 97200 \kappa_{15}^{(4)2} \kappa_3^{(4)4} \kappa_4^{(4)2} - 851472 \kappa_{15}^{(4)5} \kappa_4^{(4)3} + 995328 \kappa_{15}^{(4)4} \kappa_3^{(4)} \kappa_4^{(4)3} - \\ & 427680 \kappa_{15}^{(4)3} \kappa_3^{(4)2} \kappa_4^{(4)3} + 64800 \kappa_{15}^{(4)2} \kappa_3^{(4)3} \kappa_4^{(4)3} + 134622 \kappa_{15}^{(4)} \kappa_3^{(4)4} \kappa_4^{(4)4} - 111780 \kappa_{15}^{(4)} \kappa_3^{(4)} \kappa_4^{(4)4} + \\ & 24300 \kappa_{15}^{(4)2} \kappa_3^{(4)2} \kappa_4^{(4)4} - 11664 \kappa_{15}^{(4)3} \kappa_3^{(4)5} + 4860 \kappa_{15}^{(4)2} \kappa_3^{(4)} \kappa_4^{(4)5} + 405 \kappa_{15}^{(4)2} \kappa_4^{(4)6} + \\ & (7776 \kappa_{15}^{(4)6} - 648 \kappa_{15}^{(4)5} \kappa_3^{(4)} - 12960 \kappa_{15}^{(4)4} \kappa_3^{(4)2} + 12096 \kappa_{15}^{(4)3} \kappa_3^{(4)3} - 4320 \kappa_{15}^{(4)2} \kappa_3^{(4)4} \kappa_3^{(4)} + 576 \kappa_{15}^{(4)} \kappa_3^{(4)5} + \\ & 41472 \kappa_{15}^{(4)5} \kappa_4^{(4)} - 63828 \kappa_{15}^{(4)4} \kappa_3^{(4)} \kappa_4^{(4)} + 40176 \kappa_{15}^{(4)3} \kappa_3^{(4)2} \kappa_4^{(4)} - 12096 \kappa_{15}^{(4)2} \kappa_3^{(4)3} \kappa_4^{(4)} + 1440 \kappa_{15}^{(4)} \kappa_3^{(4)4} \kappa_4^{(4)} - \\ & 34020 \kappa_{15}^{(4)4} \kappa_4^{(4)2} + 33372 \kappa_{15}^{(4)3} \kappa_3^{(4)} \kappa_4^{(4)2} - 11664 \kappa_{15}^{(4)2} \kappa_3^{(4)2} \kappa_4^{(4)2} + 1440 \kappa_{15}^{(4)} \kappa_3^{(4)3} \kappa_4^{(4)2} + 8154 \kappa_{15}^{(4)} \kappa_3^{(4)4} \kappa_4^{(4)3} - \\ & 4752 \kappa_{15}^{(4)2} \kappa_3^{(4)} \kappa_4^{(4)3} + 720 \kappa_{15}^{(4)} \kappa_3^{(4)2} \kappa_4^{(4)3} - 702 \kappa_{15}^{(4)} \kappa_3^{(4)2} \kappa_4^{(4)4} + 180 \kappa_{15}^{(4)} \kappa_3^{(4)} \kappa_4^{(4)4} + 18 \kappa_{15}^{(4)} \kappa_4^{(4)5}) \#1 + \\ & (-7137 \kappa_{15}^{(4)4} + 7452 \kappa_{15}^{(4)3} \kappa_3^{(4)} - 3468 \kappa_{15}^{(4)2} \kappa_3^{(4)2} + 864 \kappa_{15}^{(4)} \kappa_3^{(4)3} - 112 \kappa_3^{(4)4} + 6300 \kappa_{15}^{(4)3} \kappa_4^{(4)} - \\ & 4764 \kappa_{15}^{(4)2} \kappa_3^{(4)} \kappa_4^{(4)} + 1608 \kappa_{15}^{(4)} \kappa_3^{(4)2} \kappa_4^{(4)} - 224 \kappa_3^{(4)3} \kappa_4^{(4)} - 2010 \kappa_{15}^{(4)2} \kappa_4^{(4)2} + \\ & 960 \kappa_{15}^{(4)} \kappa_3^{(4)} \kappa_4^{(4)2} - 168 \kappa_3^{(4)2} \kappa_4^{(4)2} + 186 \kappa_{15}^{(4)} \kappa_4^{(4)3} - 56 \kappa_3^{(4)} \kappa_4^{(4)3} - 7 \kappa_4^{(4)4}) \#1^2 + \\ & (-48 \kappa_{15}^{(4)2} + 36 \kappa_{15}^{(4)} \kappa_3^{(4)} - 8 \kappa_3^{(4)2} + 24 \kappa_{15}^{(4)} \kappa_4^{(4)} - 8 \kappa_3^{(4)} \kappa_4^{(4)} - 2 \kappa_4^{(4)2}) \#1^3 + \#1^4 \&, 4]/ \\ & (\kappa_{15}^{(4)2} (24 \kappa_{15}^{(4)} + (2 \kappa_3^{(4)} + \kappa_4^{(4)})^2 - 6 \kappa_{15}^{(4)} (3 \kappa_3^{(4)} + 2 \kappa_4^{(4)}))) \end{aligned} $
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	1	$-\frac{\binom{(2)}{\kappa_3}}{\binom{(4)}{\kappa_1 + \kappa_2}}$	$-\frac{2}{3(\kappa_1 + \kappa_2)}$
	3	$\frac{2\binom{(2)}{\kappa_3}}{3\binom{(4)}{\kappa_1 - 2} - 2\binom{(4)}{\kappa_2}}$	$-\frac{4}{9\binom{(4)}{\kappa_1 - 6}\binom{(4)}{\kappa_2}}$